

ANALIZAR UNA VIGA SIMPLEMENTE APOYADA BAJO LAS SIGUIENTES CONDICIONES

$$f'_c = 210 \frac{kg}{cm^2} \quad \text{Resistencia del hormigon}$$

$$f_y = 5000 \frac{kg}{cm^2} \quad \text{Limite de fluencia del acero}$$

$$E = 217370 \frac{kg}{cm^2} \quad \text{Modulo de elasticidad del hormigon}$$

$$b = 40 \text{ cm} \quad \text{Base de la viga}$$

$$H = 120 \text{ cm} \quad \text{Peralte de la viga}$$

$$c = \frac{H}{2} = 0.6 \text{ m} \quad \text{Distancia del eje neutro a la fibra mas alejada de la seccion}$$

$$L = 14 \text{ m} \quad \text{Luz de la viga}$$

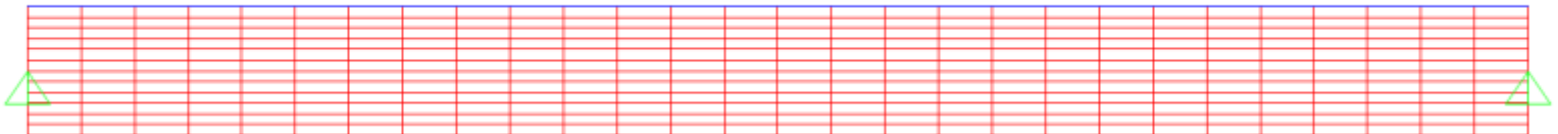
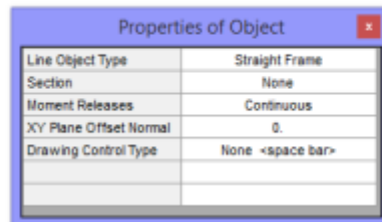
$$q = 5000 \frac{kg}{m} \quad \text{carga distribuida sobre la viga}$$

$$v = 0.2 \quad \text{coeficiente de poison}$$

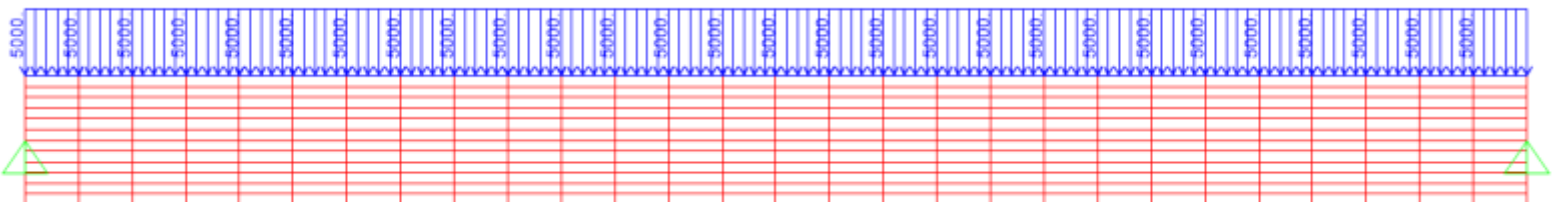
MODELADO SAP200

La estructura se modelo en el programa sap2000 con elementos Shell.

Para distribuir la carga al elemento Shell se realizó a través de un elemento frame NONE que no tiene propiedades y no aporta rigidez a la estructura, en este caso solo nos permitira la distribución de la carga hacia el elemento Shell.



Distribución de carga a través de elemento frame NONE sin propiedades.



RESOLUCION MATEMATICA TENSION DIRECCION X " σ_x " $\left[\frac{kg}{cm^2}\right]$

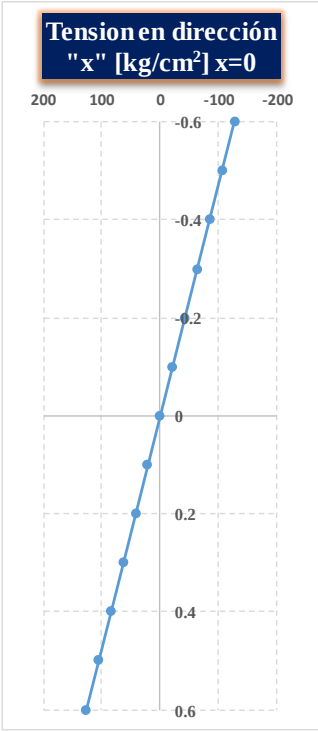
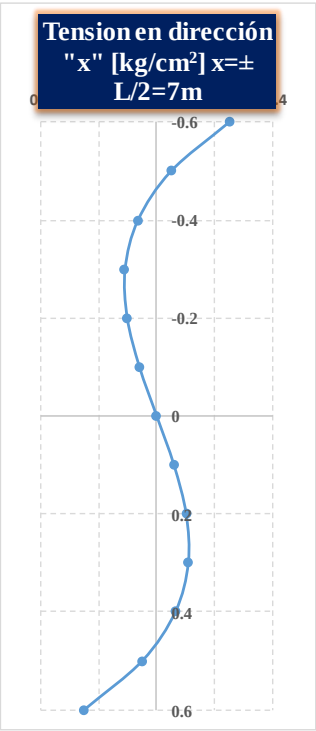
Aplicando la fórmula para tensión en dirección x, se tiene los siguientes resultados:

$$\sigma_x = \frac{q}{2I}(l^2 - x^2)y + \frac{q}{2I}\left(\frac{2}{3}y^3 - \frac{2}{5}c^2y\right)$$

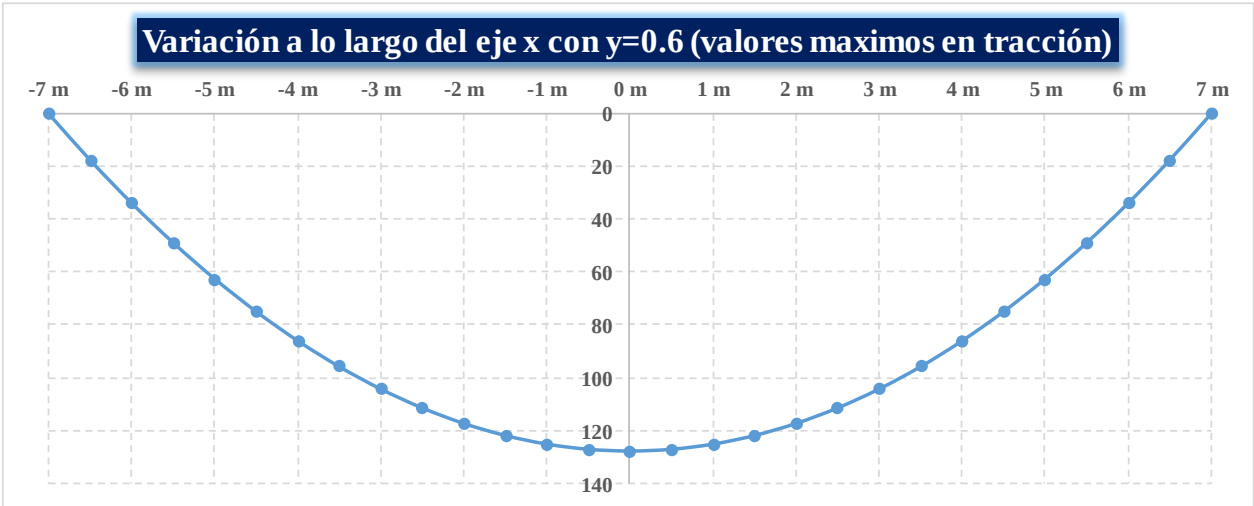
Se tiene la siguiente tabla con puntos en función a la división del elemento “Shell” expresada en unidades de $\left[\frac{kg}{cm^2}\right]$ unidades que nos permiten identificar rápidamente que puntos exceden a la resistencia característica de $f'_c = 210 \frac{kg}{cm^2}$

		Eje x [m]																															
		-7	-6.5	-6	-5.5	-5	-4.5	-4	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6	6.5	7			
Eje y [m]	-0.6	-0.250	-17.828	-34.104	-49.078	-62.750	-75.120	-86.188	-95.953	-104.417	-111.578	-117.438	-121.995	-125.250	-127.203	-127.854	-127.203	-125.250	-121.995	-117.438	-111.578	-104.417	-95.953	-86.188	-75.120	-62.750	-49.078	-34.104	-17.828	-0.250			
	-0.5	-0.049	-14.698	-28.261	-40.739	-52.133	-62.441	-71.664	-79.802	-86.855	-92.823	-97.705	-101.503	-104.216	-105.843	-106.386	-105.843	-104.216	-101.503	-97.705	-92.823	-86.855	-79.802	-71.664	-62.441	-52.133	-40.739	-28.261	-14.698	-0.049			
	-0.4	0.065	-11.654	-22.505	-32.487	-41.602	-49.848	-57.227	-63.737	-69.380	-74.154	-78.060	-81.098	-83.269	-84.571	-85.005	-84.571	-83.269	-81.098	-78.060	-74.154	-69.380	-63.737	-57.227	-49.848	-41.602	-32.487	-22.505	-11.654	0.065			
	-0.3	0.109	-8.680	-16.818	-24.305	-31.141	-37.326	-42.859	-47.742	-51.974	-55.555	-58.484	-60.763	-62.391	-63.367	-63.693	-63.367	-62.391	-60.763	-58.484	-55.555	-51.974	-47.742	-42.859	-37.326	-31.141	-24.305	-16.818	-8.680	0.109			
	-0.2	0.102	-5.758	-11.183	-16.174	-20.731	-24.855	-28.544	-31.799	-34.620	-37.008	-38.961	-40.480	-41.565	-42.216	-42.433	-42.216	-41.565	-40.480	-38.961	-37.008	-34.620	-31.799	-28.544	-24.855	-20.731	-16.174	-11.183	-5.758	0.102			
	-0.1	0.060	-2.870	-5.583	-8.078	-10.357	-12.419	-14.263	-15.891	-17.302	-18.495	-19.472	-20.231	-20.774	-21.099	-21.208	-21.099	-20.774	-20.231	-19.472	-18.495	-17.302	-15.891	-14.263	-12.419	-10.357	-8.078	-5.583	-2.870	0.060			
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
	0.1	-0.060	2.870	5.583	8.078	10.357	12.419	14.263	15.891	17.302	18.495	19.472	20.231	20.774	21.099	21.208	21.099	20.774	20.231	19.472	18.495	17.302	15.891	14.263	12.419	10.357	8.078	5.583	2.870	-0.060			
	0.2	-0.102	5.758	11.183	16.174	20.731	24.855	28.544	31.799	34.620	37.008	38.961	40.480	41.565	42.216	42.433	42.216	41.565	40.480	38.961	37.008	34.620	31.799	28.544	24.855	20.731	16.174	11.183	5.758	-0.102			
	0.3	-0.109	8.680	16.818	24.305	31.141	37.326	42.859	47.742	51.974	55.555	58.484	60.763	62.391	63.367	63.693	63.367	62.391	60.763	58.484	55.555	51.974	47.742	42.859	37.326	31.141	24.305	16.818	8.680	-0.109			
0.4	-0.065	11.654	22.505	32.487	41.602	49.848	57.227	63.737	69.380	74.154	78.060	81.098	83.269	84.571	85.005	84.571	83.269	81.098	78.060	74.154	69.380	63.737	57.227	49.848	41.602	32.487	22.505	11.654	-0.065				
0.5	0.049	14.698	28.261	40.739	52.133	62.441	71.664	79.802	86.855	92.823	97.705	101.503	104.216	105.843	106.386	105.843	104.216	101.503	97.705	92.823	86.855	79.802	71.664	62.441	52.133	40.739	28.261	14.698	0.049				
0.6	0.250	17.828	34.104	49.078	62.750	75.120	86.188	95.953	104.417	111.578	117.438	121.995	125.250	127.203	127.854	127.203	125.250	121.995	117.438	111.578	104.417	95.953	86.188	75.120	62.750	49.078	34.104	17.828	0.250				

Como se puede observar no hay un esfuerzo que supere a $f'_c = 210 \frac{kg}{cm^2}$



Se puede notar que a $x = \pm l$ se tiene una distribución de esfuerzos normales, esto es debido a la corrección dada por la segunda expresión de la fórmula de $\sigma_x(x, y)$.



VERIFICACION CON MODELO SAP2000

		Eje x [m]																													
		-7	-6.5	-6	-5.5	-5	-4.5	-4	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6	6.5	7	
Eje y [m]	-0.6	-0.250	-17.828	-34.104	-49.078	-62.750	B	-86.188	-95.953	-104.417	-111.578	-117.438	-121.995	-125.250	-127.203	-127.854	-127.203	-125.250	-121.995	-117.438	-111.578	-104.417	-95.953	-86.188	-75.120	-62.750	-49.078	-34.104	-17.828	-0.250	
	-0.5	-0.049	-14.698	-28.261	-40.739	-52.133		-71.664	-79.802	-86.855	-92.823	-97.705	-101.503	-104.216	-105.843	-106.386	-105.843	-104.216	-101.503	-97.705	-92.823	-86.855	-79.802	-71.664	-62.441	-52.133	-40.739	-28.261	-14.698	-0.049	
	-0.4	0.065	-11.654	-22.505	-32.487	-41.602	C	-57.227	-69.380	-74.154	-78.060	-81.098	-83.269	-84.571	-85.005	-84.571	-83.269	-81.098	-78.060	-74.154	-69.380	-63.737	-57.227	-49.848	-41.602	-32.487	-22.505	-11.654	0.065		
	-0.3	0.109	-8.680	-16.818	-24.305	-31.141		-37.326	-42.859	-47.742	-51.974	-55.555	-58.484	-60.763	-62.391	-63.367	-63.693	-63.367	-62.391	-60.763	-58.484	-55.555	-51.974	-47.742	-42.859	-37.326	-31.141	-24.305	-16.818	-8.680	0.109
	-0.2	0.102	-5.758	-11.183	-16.174	-20.731	-24.855	-28.544	A	-31.799	-34.620	-37.008	-38.961	-40.480	-41.565	-42.216	-42.433	-42.216	-41.565	-40.480	-38.961	-37.008	-34.620	-31.799	-28.544	-24.855	-20.731	-16.174	-11.183	-5.758	0.102
	-0.1	0.060	-2.870	-5.583	-8.078	-10.357	-12.419	-14.263		-15.891	-17.302	-18.495	-19.472	-20.231	-20.774	-21.099	-21.208	-21.099	-20.774	-20.231	-19.472	-18.495	-17.302	-15.891	-14.263	-12.419	-10.357	-8.078	-5.583	-2.870	0.060
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	0.1	-0.060	2.870	5.583	8.078	10.357	12.419	14.263	15.891	17.302	18.495	19.472	20.231	20.774	21.099	21.208	21.099	20.774	20.231	19.472	18.495	17.302	15.891	14.263	12.419	10.357	8.078	5.583	2.870	-0.060	
0.2	-0.102	5.758	11.183	16.174	20.731	24.855	28.544	31.799	34.620	37.008	38.961	40.480	41.565	42.216	42.433	42.216	41.565	40.480	38.961	37.008	34.620	31.799	28.544	24.855	20.731	16.174	11.183	5.758	-0.102		
0.3	-0.109	8.680	16.818	24.305	31.141	37.326	42.859	47.742	51.974	55.555	58.484	60.763	62.391	63.367	A	63.693	62.391	60.763	58.484	55.555	51.974	47.742	42.859	37.326	31.141	24.305	16.818	8.680	-0.109		
0.4	-0.065	11.654	22.505	32.487	41.602	49.848	57.227	63.737	69.380	74.154	78.060	81.098	83.269	84.571		84.571	83.269	81.098	78.060	74.154	69.380	63.737	57.227	49.848	41.602	32.487	22.505	11.654	-0.065		
0.5	0.049	14.698	28.261	40.739	52.133	62.441	71.664	79.802	86.855	92.823	97.705	101.503	104.216	105.843		106.386	105.843	104.216	101.503	97.705	92.823	86.855	79.802	71.664	62.441	52.133	40.739	28.261	14.698	0.049	
0.6	0.250	17.828	34.104	49.078	62.750	75.120	86.188	95.953	104.417	111.578	117.438	121.995	125.250	127.203		127.854	127.203	125.250	121.995	117.438	111.578	104.417	95.953	86.188	75.120	62.750	49.078	34.104	17.828	0.250	

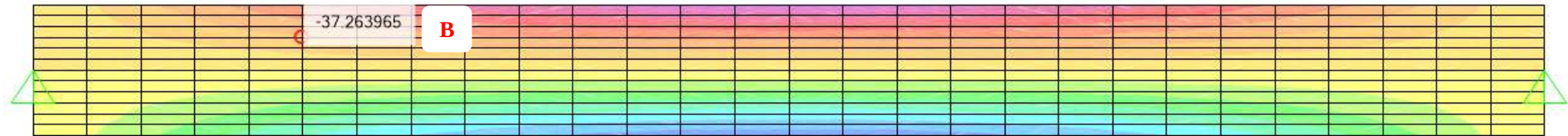
De acuerdo a la tabla anterior el punto “A” con coordenadas $x = 0m$ $y = 0.6m$ se tiene: $\sigma_x = 127.854 \frac{kg}{cm^2}$

Comparando con los resultados con el programa SAP2000 se tiene:



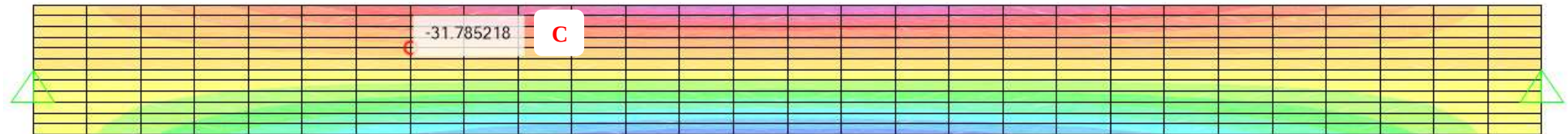
Para punto “B” con coordenadas $x = -4.5m$ $y = -0.3m$ se tiene: $\sigma_x = -37.326 \frac{kg}{cm^2}$

Con el programa SAP2000 se tiene:



Para punto “C” con coordenadas $x = -3.5m$ $y = -0.2m$ se tiene: $\sigma_x = -31.799 \frac{kg}{cm^2}$

Con el programa SAP2000 se tiene:



DISCUSIÓN DE RESULTADOS

Se puede notar que a $x = \pm 7m$ existen valores de esfuerzos por ende también existirá momentos, se sabe los momentos son nulos en los extremos de una viga apoyada simple. Estos valores esfuerzo son debido a la corrección dado por la segunda expresión de la formula.

$$\sigma_x = \frac{q}{2I}(l^2 - x^2)y + \frac{q}{2I}\left(\frac{2}{3}y^3 - \frac{2}{5}c^2y\right)$$

La expresión anterior es una solución exacta.

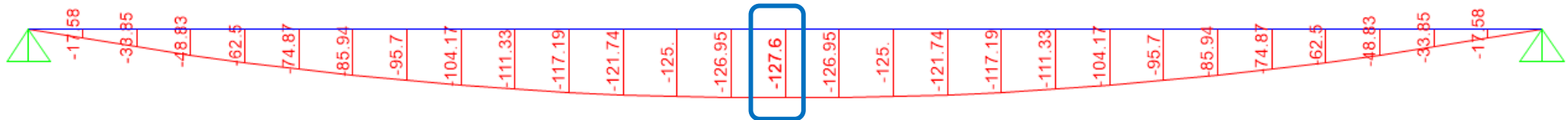
Realizando una comparación con los valores sin corregir se tiene la siguiente tabla:

		Eje x [m]																													
		-7	-6.5	-6	-5.5	-5	-4.5	-4	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6	6.5	7	
Eje y [m]	-0.6	0	-17.578	-33.854	-48.828	-62.500	-74.870	-85.938	-95.703	-104.167	-111.328	-117.188	-121.745	-125.000	-126.953	-127.604	-126.953	-125.000	-121.745	-117.188	-111.328	-104.167	-95.703	-85.938	-74.870	-62.500	-48.828	-33.854	-17.578	0	
	-0.5	0	-14.648	-28.212	-40.690	-52.083	-62.391	-71.615	-79.753	-86.806	-92.773	-97.656	-101.454	-104.167	-105.794	-106.337	-105.794	-104.167	-101.454	-97.656	-92.773	-86.806	-79.753	-71.615	-62.391	-52.083	-40.690	-28.212	-14.648	0	
	-0.4	0	-11.719	-22.569	-32.552	-41.667	-49.913	-57.292	-63.802	-69.444	-74.219	-78.125	-81.163	-83.333	-84.635	-85.069	-84.635	-83.333	-81.163	-78.125	-74.219	-69.444	-63.802	-57.292	-49.913	-41.667	-32.552	-22.569	-11.719	0	
	-0.3	0	-8.789	-16.927	-24.414	-31.250	-37.435	-42.969	-47.852	-52.083	-55.664	-58.594	-60.872	-62.500	-63.477	-63.802	-63.477	-62.500	-60.872	-58.594	-55.664	-52.083	-47.852	-42.969	-37.435	-31.250	-24.414	-16.927	-8.789	0	
	-0.2	0	-5.859	-11.285	-16.276	-20.833	-24.957	-28.646	-31.901	-34.722	-37.109	-39.063	-40.582	-41.667	-42.318	-42.535	-42.318	-41.667	-40.582	-39.063	-37.109	-34.722	-31.901	-28.646	-24.957	-20.833	-16.276	-11.285	-5.859	0	
	-0.1	0	-2.930	-5.642	-8.138	-10.417	-12.478	-14.323	-15.951	-17.361	-18.555	-19.531	-20.291	-20.833	-21.159	-21.267	-21.159	-20.833	-20.291	-19.531	-18.555	-17.361	-15.951	-14.323	-12.478	-10.417	-8.138	-5.642	-2.930	0	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	0.1	0	2.930	5.642	8.138	10.417	12.478	14.323	15.951	17.361	18.555	19.531	20.291	20.833	21.159	21.267	21.159	20.833	20.291	19.531	18.555	17.361	15.951	14.323	12.478	10.417	8.138	5.642	2.930	0	
	0.2	0	5.859	11.285	16.276	20.833	24.957	28.646	31.901	34.722	37.109	39.063	40.582	41.667	42.318	42.535	42.318	41.667	40.582	39.063	37.109	34.722	31.901	28.646	24.957	20.833	16.276	11.285	5.859	0	
	0.3	0	8.789	16.927	24.414	31.250	37.435	42.969	47.852	52.083	55.664	58.594	60.872	62.500	63.477	63.802	63.477	62.500	60.872	58.594	55.664	52.083	47.852	42.969	37.435	31.250	24.414	16.927	8.789	0	
	0.4	0	11.719	22.569	32.552	41.667	49.913	57.292	63.802	69.444	74.219	78.125	81.163	83.333	84.635	85.069	84.635	83.333	81.163	78.125	74.219	69.444	63.802	57.292	49.913	41.667	32.552	22.569	11.719	0	
	0.5	0	14.648	28.212	40.690	52.083	62.391	71.615	79.753	86.806	92.773	97.656	101.454	104.167	105.794	106.337	105.794	104.167	101.454	97.656	92.773	86.806	79.753	71.615	62.391	52.083	40.690	28.212	14.648	0	
	0.6	0	17.578	33.854	48.828	62.500	74.870	85.938	95.703	104.167	111.328	117.188	121.745	125.000	126.953	127.604	126.953	125.000	121.745	117.188	111.328	104.167	95.703	85.938	74.870	62.500	48.828	33.854	17.578	0	

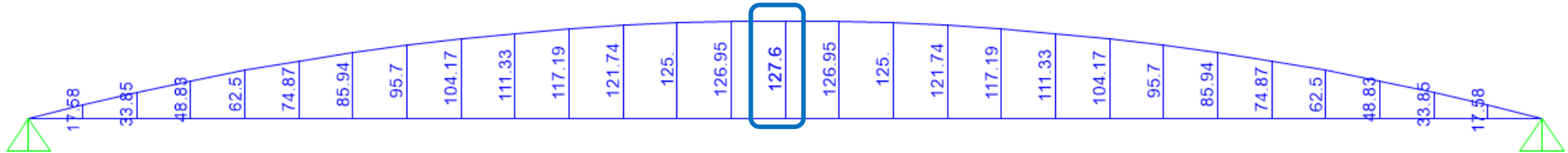
Se puede notar que no es mucha la diferencia, esto debido a que mayor longitud de la viga en comparación con la altura de la sección la corrección es insignificante.

Estos valores son las mismas obtenidas tras el análisis de un elemento frame considerando el valor máximo a $\pm 0.6m$ (análisis SAP2000)

Compresión fibra superior



Tracción fibras inferior



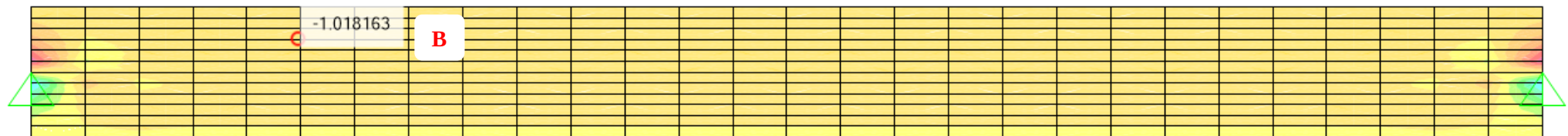
$$\sigma_{xmax} = \frac{M \cdot y}{I} \quad M = \frac{q \cdot L^2}{8} \text{ (carga distribuida)} \quad \sigma_x = \frac{5000 \frac{kg}{m} \cdot (14m)^2}{8} \cdot \frac{0.6m}{\frac{0.40 \cdot 1.2^3}{12} m^4} \cdot \left(\frac{1m}{100cm}\right)^2 \quad \sigma_x = 127.604 \frac{kg}{cm^2}$$

$$\sigma_y = -\frac{q}{2I} \left(\frac{1}{3} y^3 - c^2 y + \frac{2}{3} c^3 \right)$$

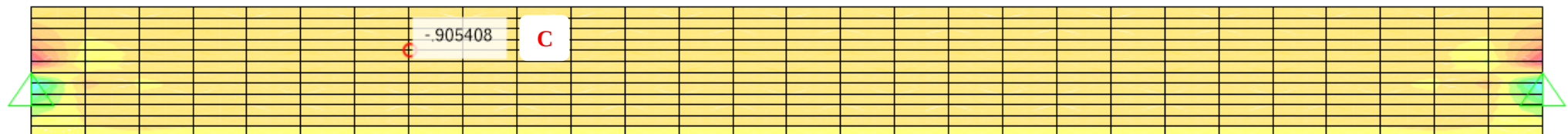
VERIFICACION CON MODELO SAP2000

$$\therefore \sigma_y = 0 \frac{kg}{cm^2}$$

Con el programa SAP2000 se tiene:



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RESOLUCION MATEMATICA DE ESFUERZO CORTANTE " τ_{xy} " $\left[\frac{kg}{cm^2} \right]$

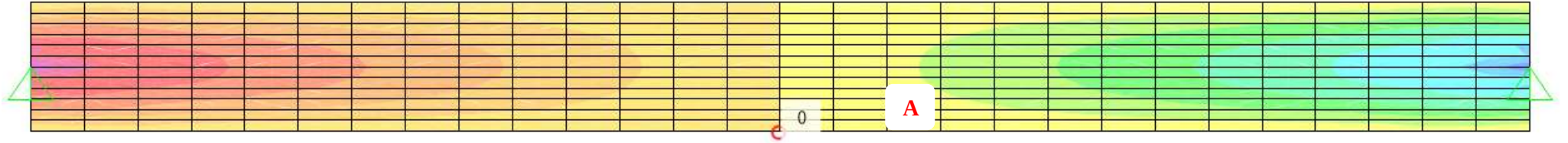
$$\tau_{xy} = -\frac{q}{2I}(c^2 - y^2)x$$

		Eje x [m]																												
		-7	-6.5	-6	-5.5	-5	-4.5	-4	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6	6.5	7
Eje y [m]	-0.6	0	0	0	0	0	B	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	-0.5	3.3420	3.1033	2.8646	2.6259	2.3872		1.9097	1.4323	1.1936	0.9549	0.7161	0.4774	0.2387	0	-0.2387	-0.4774	-0.7161	-0.9549	-1.1936	-1.4323	-1.6710	-1.9097	-2.1484	-2.3872	-2.6259	-2.8646	-3.1033	-3.3420	
	-0.4	6.0764	5.6424	5.2083	4.7743	4.3403	3.4722	C	2.6042	2.1701	1.7361	1.3021	0.8681	0.4340	0	-0.4340	-0.8681	-1.3021	-1.7361	-2.1701	-2.6042	-3.0382	-3.4722	-3.9063	-4.3403	-4.7743	-5.2083	-5.6424	-6.0764	
	-0.3	8.2031	7.6172	7.0313	6.4453	5.8594	5.2734		4.6875	3.5156	2.9297	2.3438	1.7578	1.1719	0.5859	0	-0.5859	-1.1719	-1.7578	-2.3438	-2.9297	-3.5156	-4.1016	-4.6875	-5.2734	-5.8594	-6.4453	-7.0313	-7.6172	-8.2031
	-0.2	9.7222	9.0278	8.3333	7.6389	6.9444	6.2500	5.5556	4.8611	4.1667	3.4722	2.7778	2.0833	1.3889	0.6944	0	-0.6944	-1.3889	-2.0833	-2.7778	-3.4722	-4.1667	-4.8611	-5.5556	-6.2500	-6.9444	-7.6389	-8.3333	-9.0278	-9.7222
	-0.1	10.6337	9.8741	9.1146	8.3550	7.5955	6.8359	6.0764	5.3168	4.5573	3.7977	3.0382	2.2786	1.5191	0.7595	0	-0.7595	-1.5191	-2.2786	-3.0382	-3.7977	-4.5573	-5.3168	-6.0764	-6.8359	-7.5955	-8.3550	-9.1146	-9.8741	-10.6337
	0	10.9375	0.1563	9.3750	8.5938	7.8125	7.0313	6.2500	5.4688	4.6875	3.9063	3.1250	2.3438	1.5625	0.7813	0	-0.7813	-1.5625	-2.3438	-3.1250	-3.9063	-4.6875	-5.4688	-6.2500	-7.0313	-7.8125	-8.5938	-9.3750	-10.1563	-10.9375
	0.1	10.6337	9.8741	9.1146	8.3550	7.5955	6.8359	6.0764	5.3168	4.5573	3.7977	3.0382	2.2786	1.5191	0.7595	0	-0.7595	-1.5191	-2.2786	-3.0382	-3.7977	-4.5573	-5.3168	-6.0764	-6.8359	-7.5955	-8.3550	-9.1146	-9.8741	-10.6337
	0.2	9.7222	9.0278	8.3333	7.6389	6.9444	6.2500	5.5556	4.8611	4.1667	3.4722	2.7778	2.0833	1.3889	0.6944	0	-0.6944	-1.3889	-2.0833	-2.7778	-3.4722	-4.1667	-4.8611	-5.5556	-6.2500	-6.9444	-7.6389	-8.3333	-9.0278	-9.7222
	0.3	8.2031	7.6172	7.0313	6.4453	5.8594	5.2734	4.6875	4.1016	3.5156	2.9297	2.3438	1.7578	1.1719	0.5859	0	-0.5859	-1.1719	-1.7578	-2.3438	-2.9297	-3.5156	-4.1016	-4.6875	-5.2734	-5.8594	-6.4453	-7.0313	-7.6172	-8.2031
	0.4	6.0764	5.6424	5.2083	4.7743	4.3403	3.9063	3.4722	3.0382	2.6042	2.1701	1.7361	1.3021	0.8681	0.4340	A	-0.4340	-0.8681	-1.3021	-1.7361	-2.1701	-2.6042	-3.0382	-3.4722	-3.9063	-4.3403	-4.7743	-5.2083	-5.6424	-6.0764
	0.5	3.3420	3.1033	2.8646	2.6259	2.3872	2.1484	1.9097	1.6710	1.4323	1.1936	0.9549	0.7161	0.4774	0.2387		-0.2387	-0.4774	-0.7161	-0.9549	-1.1936	-1.4323	-1.6710	-1.9097	-2.1484	-2.3872	-2.6259	-2.8646	-3.1033	-3.3420
	0.6	0	0	0	0	0	0	0	0	0	0	0	0	0	0		0	0	0	0	0	0	0	0	0	0	0	0	0	0

VERIFICACION CON MODELO SAP2000

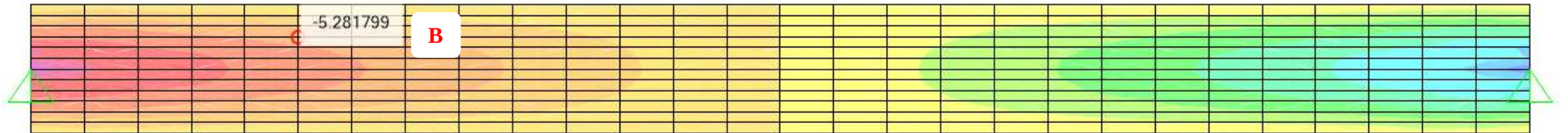
De acuerdo a la tabla anterior el punto "A" con coordenadas $x = 0m$ $y = 0.6m$ se tiene: $\tau_{xy} = 0 \frac{kg}{cm^2}$

Comparando con los resultados con el programa SAP2000 se tiene:



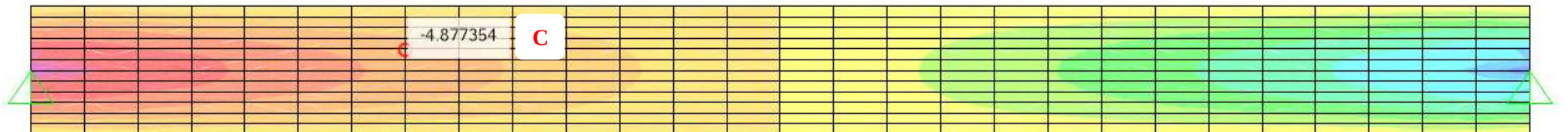
Para punto "B" con coordenadas $x = -4.5m$ $y = -0.3m$ se tiene: $\tau_{xy} = 5.2734 \frac{kg}{cm^2}$

Con el programa SAP2000 se tiene:



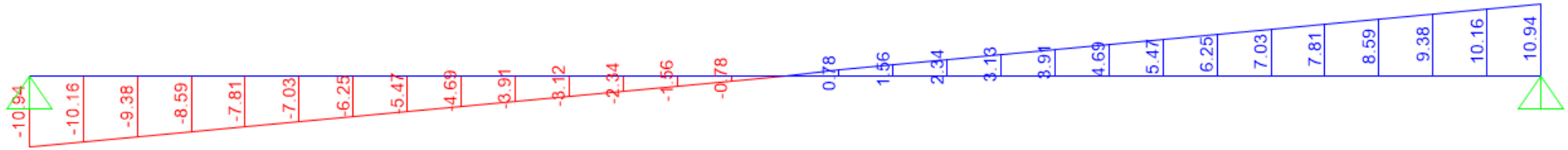
Para punto "C" con coordenadas $x = -3.5m$ $y = -0.2m$ se tiene: $\tau_{xy} = 4.8611 \frac{kg}{cm^2}$

Con el programa SAP2000 se tiene:



DISCUSIÓN DE RESULTADOS

Realizando una comparación con los esfuerzos cortantes en elementos frame:



Se puede notar que son los mismos resultados obtenidos para $y=0$ en la tabla anterior

Utilizando la formula conocida del esfuerzo cortante para sección rectangular

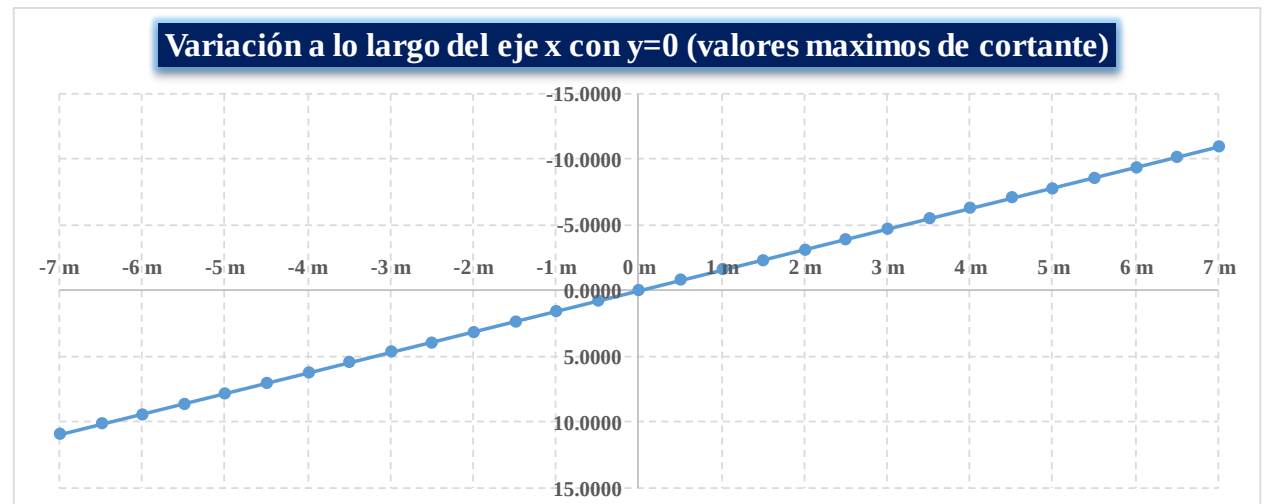
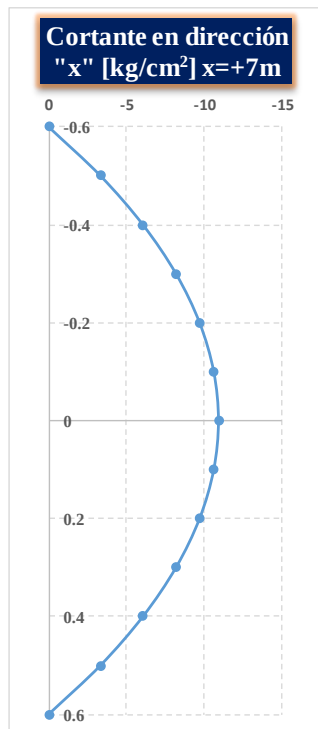
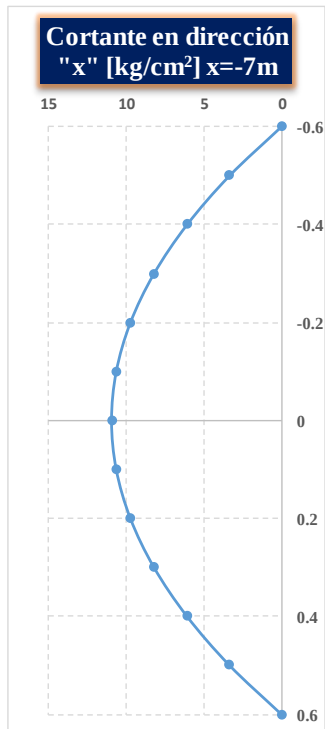
$$\tau_{xy} = \frac{V}{2I} \left(\frac{H^2}{4} - y^2 \right)$$

Cortantes máximos en los apoyos: $x = \pm 7m$; $y = 0$

Que se verifica con la formula $\tau_{xy}(x, y)$.

$$\tau_{xy} = \pm \frac{\frac{5000 \frac{kg}{m} \cdot 14m}{2} \cdot \frac{1.2m^2}{4} - 0^2}{2 \cdot \frac{0.40 \cdot 1.2^3}{12} m^4} \cdot \left(\frac{1m}{100cm} \right)^2$$

$$\tau_{xy} = \pm 10.9375 \frac{kg}{cm^2}$$



$$u = \frac{q}{2EI} \left[\left(l^2 x - \frac{x^3}{3} \right) y + x \left(\frac{2}{3} y^3 - \frac{2}{5} c^2 y \right) + v x \left(\frac{1}{3} y^3 - c^2 y + \frac{2}{3} c^3 \right) \right]$$

$$v = - \frac{q}{2EI} \left\{ \frac{y^4}{12} - \frac{c^2 y^2}{2} + \frac{2}{3} c^3 y + v \left[(l^2 - x^2) \frac{y^2}{2} + \frac{y^4}{6} - \frac{1}{5} c^2 y^2 \right] \right\}$$

$$- \frac{q}{2EI} \left[\frac{l^2 x^2}{2} - \frac{x^4}{12} - \frac{1}{5} c^2 x^2 + \left(1 + \frac{1}{2} v \right) c^2 x^2 \right] + \delta$$

$$\delta = \frac{5}{24} \frac{ql^4}{EI} \left[1 + \frac{12}{5} \frac{c^2}{l^2} \left(\frac{4}{5} + \frac{v}{2} \right) \right]$$

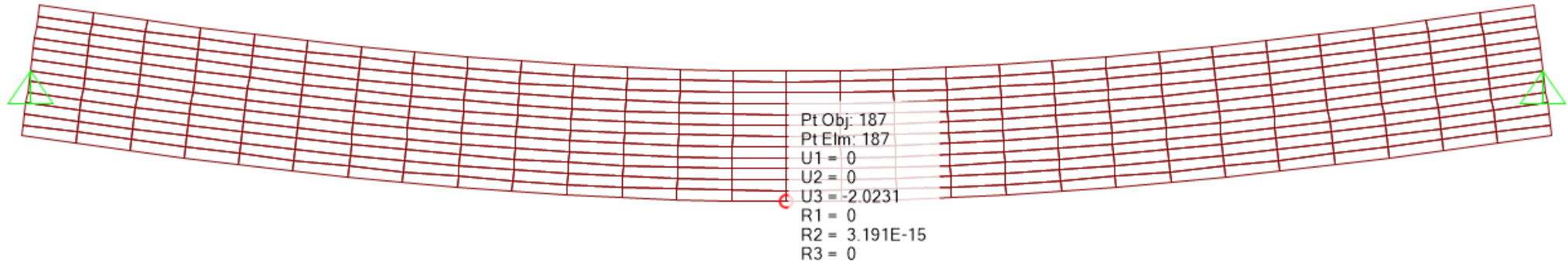
		Eje x [m]																													
		-7	-6.5	-6	-5.5	-5	-4.5	-4	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6	6.5	7	
Eje y [m]	-0.6	0	0.27190	0.26596	0.25643	0.24360	0.20926	0.18834	0.16533	0.14052	0.11421	0.08671	0.05830	0.02930	0	-0.02930	-0.05830	-0.08671	-0.11421	-0.14052	-0.16533	-0.18834	-0.20926	-0.22778	-0.24360	-0.25643	-0.26596	-0.27190	0		
	-0.5	0	0.22600	0.22110	0.21320	0.20255	0.17402	0.13750	0.11687	0.09500	0.07212	0.04850	0.02437	0	-0.02437	-0.04850	-0.07212	-0.09500	-0.11687	-0.13750	-0.15664	-0.17402	-0.18941	-0.20255	-0.21320	-0.22110	-0.22600	0			
	-0.4	0	0.18038	0.17649	0.17020	0.16172	0.13896	0.10981	0.09334	0.07587	0.05760	0.03873	0.01947	0	-0.01947	-0.03873	-0.05760	-0.07587	-0.09334	-0.10981	-0.12508	-0.13896	-0.15124	-0.16172	-0.17020	-0.17649	-0.18038	0			
	-0.3	0	0.13499	0.13210	0.12740	0.12100	0.11323	0.10404	0.08222	0.06989	0.05681	0.04313	0.02900	0.01458	0	-0.01458	-0.02900	-0.04313	-0.05681	-0.06989	-0.08222	-0.09365	-0.10404	-0.11323	-0.12100	-0.12740	-0.13210	-0.13499	0		
	-0.2	0	0.08978	0.08786	0.08475	0.08054	0.07533	0.06927	0.06232	0.05471	0.04651	0.03781	0.02870	0.01930	0.00970	0	-0.00970	-0.01930	-0.02870	-0.03781	-0.04651	-0.05471	-0.06232	-0.06922	-0.07533	-0.08054	-0.08475	-0.08786	-0.08978	0	
	-0.1	0	0.04467	0.04373	0.04219	0.04010	0.03752	0.03448	0.03104	0.02726	0.02317	0.01884	0.01430	0.00962	0.00483	0	-0.00483	-0.00962	-0.01430	-0.01884	-0.02317	-0.02726	-0.03104	-0.03448	-0.03752	-0.04010	-0.04219	-0.04373	-0.04467	0	
	0	0	-0.00037	-0.00035	-0.00032	-0.00029	-0.00026	-0.00023	-0.00020	-0.00017	-0.00014	-0.00012	-0.00009	-0.00006	-0.00003	0	0.00003	0.00006	0.00009	0.00012	0.00014	0.00017	0.00020	0.00023	0.00026	0.00029	0.00032	0.00035	0.00037	0	
	0.1	0	-0.04542	-0.04442	-0.04283	-0.04068	-0.03803	-0.03494	-0.03145	-0.02760	-0.02346	-0.01907	-0.01448	-0.00973	-0.00489	0	0.00489	0.00973	0.01448	0.01907	0.02346	0.02760	0.03145	0.03494	0.03803	0.04068	0.04283	0.04442	0.04542	0	
	0.2	0	-0.09052	-0.08855	-0.08538	-0.08111	-0.07585	-0.06968	-0.06272	-0.05506	-0.04680	-0.03804	-0.02888	-0.01942	-0.00976	0	0.00976	0.01942	0.02888	0.03804	0.04680	0.05506	0.06272	0.06968	0.07585	0.08111	0.08538	0.08855	0.09052	0	
	0.3	0	-0.13574	-0.13279	-0.12804	-0.12164	-0.11374	-0.10450	-0.09406	-0.08257	-0.07018	-0.05704	-0.04331	-0.02912	-0.01463	0	0.01463	0.02912	0.04331	0.05704	0.07018	0.08257	0.09406	0.10450	0.11374	0.12164	0.12804	0.13279	0.13574	0	
	0.4	0	-0.18113	-0.17718	-0.17084	-0.16229	-0.15175	-0.13942	-0.12549	-0.11016	-0.09363	-0.07610	-0.05777	-0.03885	-0.01952	0	0.01952	0.03885	0.05777	0.07610	0.09363	0.11016	0.12549	0.13942	0.15175	0.16229	0.17084	0.17718	0.18113	0	
	0.5	0	-0.22675	-0.22179	-0.21383	-0.20312	-0.18993	-0.17448	-0.15704	-0.13785	-0.11716	-0.09523	-0.07229	-0.04861	-0.02443	0	0.02443	0.04861	0.07229	0.09523	0.11716	0.13785	0.15704	0.17448	0.18993	0.20312	0.21383	0.22179	0.22675	0	
	0.6	0	-0.27265	-0.26665	-0.25706	-0.24418	-0.22829	-0.20972	-0.18874	-0.16567	-0.14081	-0.11444	-0.08688	-0.05842	-0.02936	0	0.02936	0.05842	0.08688	0.11444	0.14081	0.16567	0.18874	0.20972	0.22829	0.24418	0.25706	0.26665	0.27265	0	

		Eje x [m]																												
		-7	-6.5	-6	-5.5	-5	-4.5	-4	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6	6.5	7
Eje y [m]	-0.6	0	0.23188	0.45985	0.68106	0.89264	B	1.27670	1.44467	1.59404	1.72317	1.83070	1.91549	1.97669	2.01365	2.02602	2.01365	1.97669	1.91549	1.83070	1.72317	1.59404	1.44467	1.27670	1.09198	0.89264	0.68106	0.45985	0.23188	0
	-0.5	0	0.23198	0.46008	0.68141	0.89311		1.27736	1.59486	1.72406	1.83163	1.91647	1.97769	2.01467	2.02704	2.01467	1.97769	1.91647	1.83163	1.72406	1.59486	1.44542	1.27736	1.09255	0.89311	0.68141	0.46008	0.23198	0	
	-0.4	0	0.23204	0.46026	0.68170	0.89349	C	1.27790	1.59553	1.72477	1.83238	1.91725	1.97849	2.01549	2.02786	2.01549	1.97849	1.91725	1.83238	1.72477	1.59553	1.44603	1.27790	1.09301	0.89349	0.68170	0.46026	0.23204	0	
	-0.3	0	0.23208	0.46039	0.68191	0.89377		1.27831	1.59603	1.72531	1.83296	1.91785	1.97911	2.01612	2.02849	2.01612	1.97911	1.91785	1.83296	1.72531	1.59603	1.44649	1.27831	1.09336	0.89377	0.68191	0.46039	0.23208	0	
	-0.2	0	0.23211	0.46047	0.68205	0.89397	1.27851	1.59639	1.72569	1.83336	1.91827	1.97954	2.01656	2.02894	2.01656	1.97954	1.91827	1.83336	1.72569	1.59639	1.44681	1.27859	1.09361	0.89397	0.68205	0.46047	0.23211	0		
	-0.1	0	0.23211	0.46051	0.68212	0.89407	1.27875	1.59659	1.72591	1.83359	1.91851	1.97979	2.01681	2.02919	2.01681	1.97979	1.91851	1.83359	1.72591	1.59659	1.44699	1.27875	1.09374	0.89407	0.68212	0.46051	0.23211	0		
	0	0	0.23209	0.46050	0.68212	0.89408	1.27879	1.59663	1.72596	1.83365	1.91857	1.97986	2.01687	2.02926	2.01687	1.97986	1.91857	1.83365	1.72596	1.59663	1.44703	1.27879	1.09376	0.89408	0.68212	0.46050	0.23209	0		
	0.1	0	0.23205	0.46045	0.68206	0.89401	1.27868	1.59653	1.72585	1.83354	1.91845	1.97973	2.01675	2.02913	2.01675	1.97973	1.91845	1.83354	1.72585	1.59653	1.44693	1.27869	1.09368	0.89401	0.68206	0.46045	0.23205	0		
	0.2	0	0.23199	0.46036	0.68193	0.89385	1.27848	1.59627	1.72558	1.83325	1.91816	1.97943	2.01644	2.02882	2.01644	1.97943	1.91816	1.83325	1.72558	1.59627	1.44670	1.27848	1.09349	0.89385	0.68193	0.46036	0.23199	0		
	0.3	0	0.23191	0.46021	0.68173	0.89360	1.27814	1.59586	1.72514	1.83279	1.91768	1.97894	2.01594	A	2.01594	1.97894	1.91768	1.83279	1.72514	1.59586	1.44632	1.27814	1.09319	0.89360	0.68173	0.46021	0.23191	0		
0.4	0	0.23181	0.46003	0.68147	0.89326	1.27767	1.59530	1.72454	1.83215	1.91702	1.97826	2.01526	2.01526		1.97826	1.91702	1.83215	1.72454	1.59530	1.44580	1.27767	1.09278	0.89326	0.68147	0.46003	0.23181	0			
0.5	0	0.23169	0.45979	0.68113	0.89283	1.27708	1.59458	1.72377	1.83134	1.91618	1.97740	2.01438	2.01438		1.97740	1.91618	1.83134	1.72377	1.59458	1.44514	1.27708	1.09227	0.89283	0.68113	0.45979	0.23169	0			
0.6	0	0.23154	0.45950	0.68071	0.89230	1.27635	1.59433	1.72330	1.83035	1.91515	1.97634	2.01331	2.02567		2.01331	1.97634	1.91515	1.83035	1.72283	1.59370	1.44433	1.27635	1.09163	0.89230	0.68071	0.45950	0.23154	0		

VERIFICACION CON MODELO SAP2000

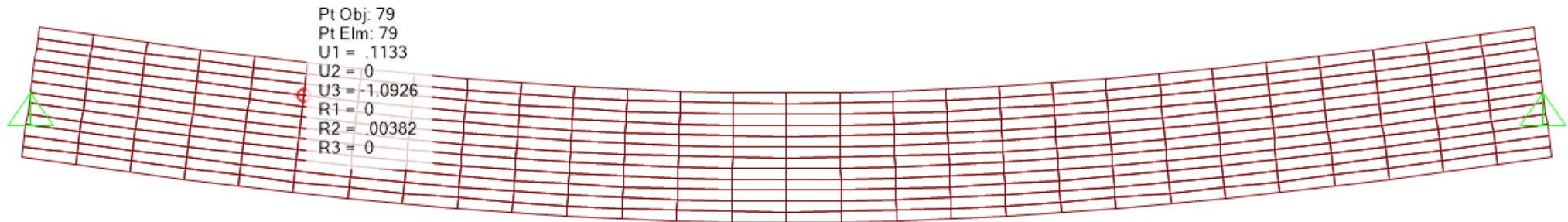
De acuerdo a la tabla anterior el punto “A” con coordenadas $x = 0m$ $y = 0.6m$ se tiene: $u = 0 \text{ cm}$ $v = 2.02567 \text{ cm}$ (positivo en dirección abajo)

Comparando con los resultados con el programa SAP2000 se tiene:



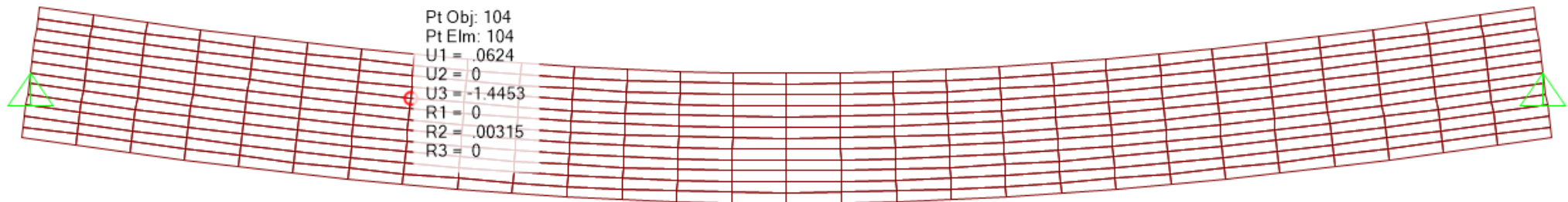
Para punto “B” con coordenadas $x = -4.5m$ $y = -0.3m$ se tiene: $u = 0.11323 \text{ cm}$ $v = 1.09336 \text{ cm}$ (positivo en dirección abajo)

Con el programa SAP2000 se tiene:

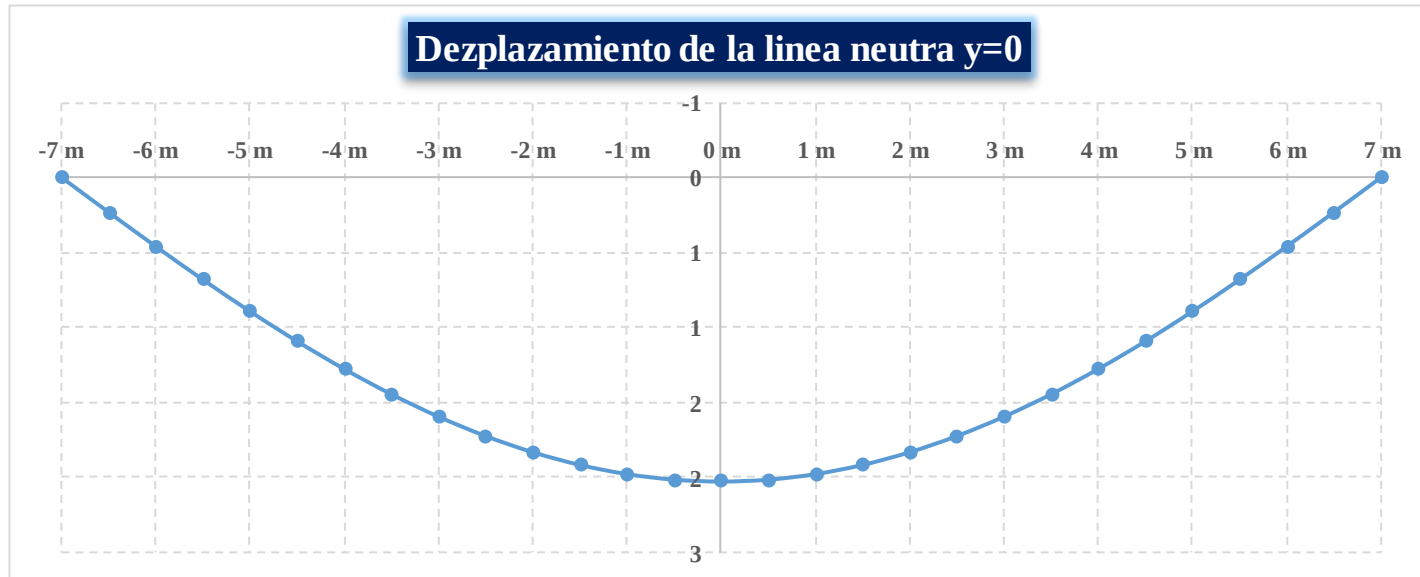


Para punto “C” con coordenadas $x = -3.5m$ $y = -0.2m$ se tiene: $u = 0.06232 \text{ cm}$ $v = 1.44681 \text{ cm}$ (positivo en dirección abajo)

Con el programa SAP2000 se tiene:



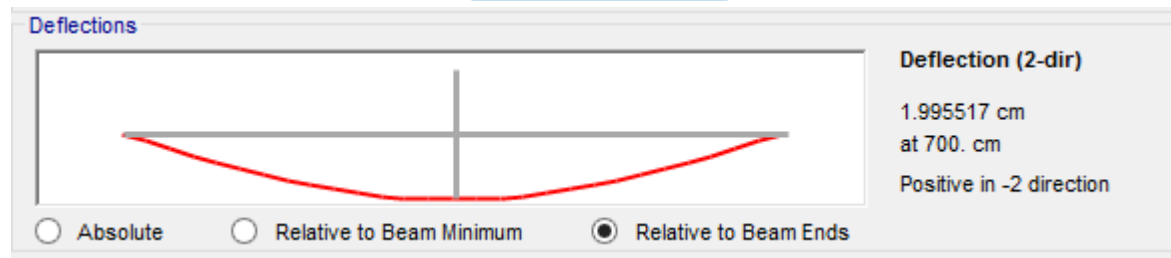
DISCUSION DE RESULTADOS



Utilizando la formula conocida

$$\delta_y = \frac{5}{24EI} q \cdot l^4 = \frac{5}{24 \cdot 217370 \frac{kg}{cm^2} \cdot \frac{0.40 \cdot 1.2^3}{12} m^4} \cdot \frac{5000 kg}{100 cm} \cdot \left(\frac{14m}{2}\right)^4$$

$$\delta_y = 1.997555 cm$$



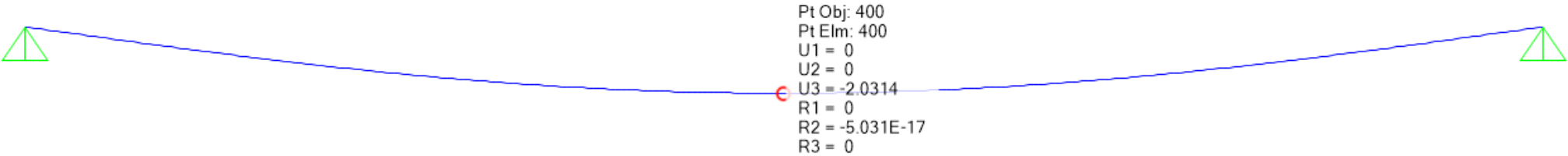
Resultado obtenido analizando un solo elemento frame verificando la fórmula de δ_y , pero no se verifica con lo obtenido de $v(x,y)$, esta variación es debido a la corrección por el efecto de fuerza cortante.

$$\delta = \frac{5}{24} \frac{ql^4}{EI} \left[1 + \frac{12}{5} \frac{c^2}{l^2} \left(\frac{4}{5} + \frac{\nu}{2} \right) \right]$$

Sin considerar la corrección se tiene la siguiente tabla, la cual verifica el resultado de δ_y

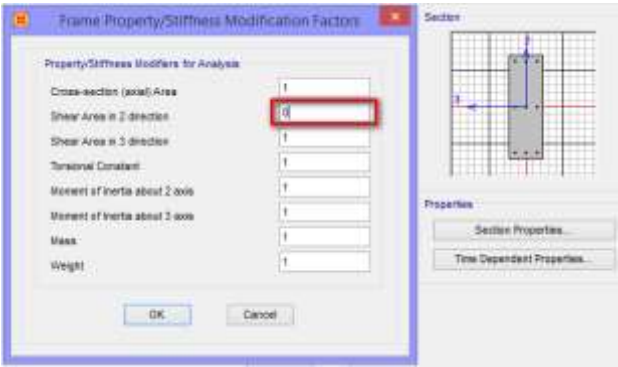
		Eje x [m]																												
		-7	-6.5	-6	-5.5	-5	-4.5	-4	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6	6.5	7
Eje y [m]	-0.6	0	0.20018	0.42815	0.64936	0.86094	1.06028	1.24500	1.41297	1.56234	1.69147	1.79900	1.88379	1.94499	1.98195	1.99432	1.98195	1.94499	1.88379	1.79900	1.69147	1.56234	1.41297	1.24500	1.06028	0.86094	0.64936	0.42815	0.20018	0
	-0.5	0	0.20028	0.42838	0.64971	0.86141	1.06085	1.24566	1.41373	1.56316	1.69236	1.79993	1.88477	1.94599	1.98297	1.99534	1.98297	1.94599	1.88477	1.79993	1.69236	1.56316	1.41373	1.24566	1.06085	0.86141	0.64971	0.42838	0.20028	0
	-0.4	0	0.20034	0.42856	0.65000	0.86179	1.06131	1.24620	1.41433	1.56383	1.69307	1.80068	1.88555	1.94679	1.98379	1.99616	1.98379	1.94679	1.88555	1.80068	1.69307	1.56383	1.41433	1.24620	1.06131	0.86179	0.65000	0.42856	0.20034	0
	-0.3	0	0.20038	0.42869	0.65021	0.86207	1.06166	1.24661	1.41479	1.56433	1.69361	1.80126	1.88615	1.94741	1.98442	1.99679	1.98442	1.94741	1.88615	1.80126	1.69361	1.56433	1.41479	1.24661	1.06166	0.86207	0.65021	0.42869	0.20038	0
	-0.2	0	0.20041	0.42877	0.65035	0.86227	1.06191	1.24689	1.41511	1.56469	1.69399	1.80166	1.88657	1.94784	1.98486	1.99724	1.98486	1.94784	1.88657	1.80166	1.69399	1.56469	1.41511	1.24689	1.06191	0.86227	0.65035	0.42877	0.20041	0
	-0.1	0	0.20041	0.42881	0.65042	0.86237	1.06204	1.24705	1.41529	1.56489	1.69421	1.80189	1.88681	1.94809	1.98511	1.99749	1.98511	1.94809	1.88681	1.80189	1.69421	1.56489	1.41529	1.24705	1.06204	0.86237	0.65042	0.42881	0.20041	0
	0	0	0.20039	0.42880	0.65042	0.86238	1.06206	1.24709	1.41533	1.56493	1.69426	1.80195	1.88687	1.94816	1.98517	1.99756	1.98517	1.94816	1.88687	1.80195	1.69426	1.56493	1.41533	1.24709	1.06206	0.86238	0.65042	0.42880	0.20039	0
	0.1	0	0.20035	0.42875	0.65036	0.86231	1.06198	1.24699	1.41523	1.56483	1.69415	1.80184	1.88675	1.94803	1.98505	1.99743	1.98505	1.94803	1.88675	1.80184	1.69415	1.56483	1.41523	1.24699	1.06198	0.86231	0.65036	0.42875	0.20035	0
	0.2	0	0.20029	0.42866	0.65023	0.86215	1.06179	1.24678	1.41500	1.56457	1.69388	1.80155	1.88646	1.94773	1.98474	1.99712	1.98474	1.94773	1.88646	1.80155	1.69388	1.56457	1.41500	1.24678	1.06179	0.86215	0.65023	0.42866	0.20029	0
	0.3	0	0.20021	0.42851	0.65003	0.86190	1.06149	1.24644	1.41462	1.56416	1.69344	1.80109	1.88598	1.94724	1.98424	1.99662	1.98424	1.94724	1.88598	1.80109	1.69344	1.56416	1.41462	1.24644	1.06149	0.86190	0.65003	0.42851	0.20021	0
0.4	0	0.20011	0.42833	0.64977	0.86156	1.06108	1.24597	1.41410	1.56360	1.69284	1.80045	1.88532	1.94656	1.98356	1.99593	1.98356	1.94656	1.88532	1.80045	1.69284	1.56360	1.41410	1.24597	1.06108	0.86156	0.64977	0.42833	0.20011	0	
0.5	0	0.19999	0.42809	0.64943	0.86113	1.06057	1.24538	1.41344	1.56288	1.69207	1.79964	1.88448	1.94570	1.98268	1.99505	1.98268	1.94570	1.88448	1.79964	1.69207	1.56288	1.41344	1.24538	1.06057	0.86113	0.64943	0.42809	0.19999	0	
0.6	0	0.19984	0.42780	0.64901	0.86060	1.05993	1.24465	1.41263	1.56200	1.69113	1.79865	1.88345	1.94464	1.98161	1.99397	1.98161	1.94464	1.88345	1.79865	1.69113	1.56200	1.41263	1.24465	1.05993	0.86060	0.64901	0.42780	0.19984	0	

Si se divide el elemento frame en varios elementos se tiene los siguientes resultados



Que verifican con los obtenidos de la formula de $v(x, y)$, esto indica que tras la división existe una interacción entre elemento que produce el efecto de cortante que el programa sap2000 lo considra para su analisis.

Se puede anular el efecto de cortante en elementos frame con los modificadores de propiedades “área de corte nulo”.



Con la cual se verifica los valores de $\delta_y = \frac{5}{24EI} q \cdot l^4$

